

CRYSTALLOGRAPHY

BY

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RESEARCH AND THEORY

A PUBLISHED CELL EDGE READ BACK OFF A VOXEL GRID, AND WHOSE RESULT
THAT IS

ORIOR

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*For Megan, my loving wife, the only person that really understands
what I mean when I speak.*

*For Autumn and Opal, without you this thought would have never
occurred.*

To everyone else, I wish you well.

ABSTRACT

A crystal carries an answer somebody else wrote down before this instrument existed. No other control in this work does.

Tile a published unit cell, carry every site as an exact integer, and score how well the arrangement agrees with itself shifted along one axis. The offset where agreement peaks is the cell edge. Across 453 axes drawn from the Crystallography Open Database, every recovered period equals the published edge as an integer, with no tolerance applied and no error left to average.

That exactness is recent and the chapter keeps what it replaced. Until 2026-09-09 the sites were assigned to voxels of 0.25 angstroms and the tiling was capped at 320 voxels along the longest side. Both bounds came from this work and neither came from the deposit, and they cost a mean absolute error of 0.0124 angstroms together with a bias of -0.0067 that the reading could not account for. The cap also refused large cells outright, which never appeared as error because those entries went unread. Carrying the sites as exact integers removes the sample spacing, the straddling lags, the fraction to resolve and the error distribution at once.

One correction here is not about the reader. A lattice of period P agrees with itself as well at $2P$ and at $3P$. Taking the tallest lag reports a harmonic. Read that way, published edges came back at almost exactly twice their value on ten of eighteen axes. Scoring a candidate together with its multiples fixes it, and that rule survives the removal of the grid unchanged, because it repairs a property of periodic sets and not damage the reader did.

What this does not license is stated with it. The reading recovers the number the deposit carries and stops there, only right-angled cells are used, and exact agreement does not make the published edge true. The interest is not the cell edge, which is published to more decimal places than anything here reaches. The interest is that a memoryless control can only show an instrument does not invent structure, and a crystal lets it show that it finds structure that is there.

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WHAT THIS IS

Purpose: State the construction, and mark the boundary of what this work can claim about it. **Scope:** `examples/crystallography/6_oracle/proof_positive_control.py`, `src/engine/python/measure/shift_agreement.py`, `src/engine/python/representation/exact.py`

Note, 26 September. `shift_agreement.py` and `exact.py` were deleted with `src/engine/python/` by commit 74601c1 on 26 September, and neither is in a file at orior 1948ae1; `proof_positive_control.py` still imports both. **Later, 26 September.** Commit d09b489 (orior PR 12) put back the 126 files 74601c1 deleted, each byte-identical to its state before the deletion, and biohub pins orior at d09b489. Every file named here that 74601c1 deleted is at the pin again; nothing here was rerun against them.

1.1 THE CONSTRUCTION

Tile a published unit cell, carry every site as an exact integer, and score how well the arrangement agrees with itself shifted by each candidate offset along one axis. The offset where agreement peaks is the cell edge.

That is an autocorrelation. `lattice_agreement` takes one lag, compares the grid against a copy of itself shifted by that lag along one axis, and returns the share of *occupied* cells whose neighbor at that distance carries the same value. The caller sweeps the lag.

Empty cells are left out. The measurement depends on that exclusion: a voxel grid built from atom sites is mostly empty. Counting the empty positions would report how empty the grid is, which every structure agrees on. The claim is checkable against `src/engine/python/measure/shift_agreement.py:97`.

1.2 SOMEBODY HAS ALMOST CERTAINLY BEEN HERE FIRST

An autocorrelation of density in a crystal is very unlikely to be new. A century of crystallography has worked on exactly this object, and the reasonable prior is that this construction has a name, a date and a literature behind it.

That is the good outcome and not the bad one. This construction was reached from a general question about how far an arrangement sits from the most disordered arrangement of its own parts, with no crystal in mind. Arriving from there at something a field already uses is evidence that the general question is pointed at something real. Arriving somewhere nobody had ever needed to go would have been the weaker result.

So no priority is claimed here, and nobody wants any. The reading is missing: somebody has to work through the crystallographic literature and write down what it says, and that has not been done. Until it is, nothing in these chapters is a claim to a new method and nothing in them is an attribution either. The reading is worth doing for its own sake, and whoever does it will find out what this actually is.

1.3 WHY THE CRYSTAL IS HERE AS A CONTROL

The interest of this chapter is not the cell edge. Cell edges are published, refined to more decimal places than anything here reaches, and recovering one adds nothing to crystallography.

The interest is that the answer is known in advance by somebody else. Every other control in this work until the protein structures was a memoryless process, and a memoryless control shows only that an instrument does not invent structure. It cannot show that the instrument finds structure that is there. Those are different claims, and an instrument has to earn the second.

A crystal earns it. The periodicity is not inferred from the measurement, it is a fact about the mineral, and the Crystallography Open Database publishes the cell for every entry it holds. The number the detector has to hit was written down before this instrument existed, by people with no interest in whether it works.

WHAT IS ACTUALLY MEASURED

Purpose: State the pipeline from a deposited structure to a recovered edge, leaving a reader able to say what the number is a number about. **Scope:** `src/engine/python/representation/exact.py`, `src/engine/python/representation/structure/crystal.py`, `src/engine/python/measure/shift_agreement.py`

Note, 26 September. The three files in this scope were deleted with `src/engine/python/` by commit 74601c1 on 26 September, and none is in a file at orior 1948ae1. This chapter states the pipeline as it was built before the deletion. **Later, 26 September.** Commit d09b489 (orior PR 12) put back the 126 files 74601c1 deleted, each byte-identical to its state before the deletion, and biohub pins orior at d09b489. Every file named here that 74601c1 deleted is at the pin again; nothing here was rerun against them.

2.1 FROM A DEPOSITED CELL TO EXACT POINTS

A CIF entry gives the cell lengths, the angles, and the fractional coordinates of the sites in one cell. Four steps turn that into something the detector reads.

1. **The angles are checked.** Only cells with right angles are used. An oblique cell would need a shear before a shift along one coordinate meant a shift along an axis, and that shear is a second thing able to go wrong between the published number and the recovered one.
2. **The cell is tiled.** A single cell holds one period and one period cannot be told from noise. Four tiles an axis, and there is no cap on this: the points are sparse. The cost is linear in the tile count and not the cube of a grid side.
3. **The sites are carried as exact integers.** A deposit writes decimal text, and that text is a numerator and a count of decimal places, both integers. A fractional coordinate plus a whole tile, times a cell edge, is a product of two decimals carried through as integers. Nothing rounds. There is no grid, no voxel, and no nearest anything.
4. **Each axis is asked separately** for its period, and all three answers are kept. One structure therefore contributes up to three independent results.

Nothing at any step tells the detector what it is looking at. It receives positions carrying values and an axis index.

This was a grid until 2026-09-09. Sites were assigned to voxels of 0.25 angstroms and the tiling was capped at 320 voxels along the longest side. Both bounds came from this work and neither came from the deposit, and what they cost is measured in the evidence chapter: a mean absolute error of 0.0124 angstroms on a measurement whose error is otherwise zero. The cap also refused large cells outright, which is a second cost that never appeared as error because those entries simply went unread.

2.2 READING AN AXIS WITHOUT AN ORDER

An axis is read as a lookup from a coordinate to the arrangement standing at it. Two coordinates agree when their arrangements match, values included, and a period is a coordinate whose arrangement matches the one exactly a period away.

The values have to be in the comparison. Leaving them out reads a rocksalt cell at half its published edge, correctly: the two sublattices interleave. The positions alone do repeat every $a/2$ and only the elements distinguish the two halves.

The candidates are every difference between two occupied coordinates, and that set is complete. A period that agrees with anything at all is by construction a difference between two things that agree. It is in the set. There is no sweep, no ceiling and no stride, and nothing in the measure narrows what can be found.

Membership is the only relation asked of the set. The search is a lookup. A binary search over sorted coordinates answers the same question and imposes an order the set does not have, paying a full comparison at every step of it. The engine's C arms did exactly that until it was noticed.

Taking the tallest lag is wrong, and it was taken that way first. A lattice of period P agrees with itself just as well at $2P$ and at $3P$. Which of those is tallest is settled by noise. The tallest lag reports a harmonic and not the fundamental. Read that way, published cell edges came back at almost exactly twice their value on ten of eighteen axes.

The correction is to score a candidate on itself together with its multiples, and to require at least two members of that family to be present before the candidate is scored. Exactly two members are used and never more: a family growing as the candidate shrinks lets a short wrong candidate win by holding more members, since it only has to catch one good lag among them. Herzenbergite, molybdite and one more each failed that way before the family was capped, reading short by a factor near two thirds.

The family rule survives the removal of the grid unchanged, and that is worth noting. Every other correction in this chapter turned out to be repairing damage the reader had done. This one repairs a property of periodic sets themselves: a set with period P agrees with itself at $2P$ whether or not anybody rounded it. The rule is needed in the exact reading exactly as it was needed in the gridded one.

This is the harmonic trap, and it is not specific to crystals. The same failure has bitten three modules in this work independently. Scoring against multiples instead of taking the tallest peak is the general fix, and the crystal case is where it is easiest to see, because the right answer is published and a factor of two is unmistakable.

2.3 WHAT A FAILURE WOULD HAVE LOOKED LIKE

Stated before the numbers, since afterwards it is worth nothing.

A recovered period not equal to the published edge would mean the detector is not reading the period. A near miss counts as a miss, since nothing in this path rounds and there is no tolerance to fall inside. A systematic bias in one direction would mean it is reading the tiling and not the crystal, since the tiling is the only structure present that the mineral did not put there. Both are visible in the reported columns without knowing which entry is which.

The gridded reading set that bar at one voxel instead, because it had a voxel. It passed 453 of 453 and carried a bias of -0.0067 angstroms that it could not account for. The exact reading passes the stricter test and has no bias to explain. That is the same evidence, read without a floor under it.

THE EVIDENCE

Purpose: Report every number the sweep produced, including the ones that went furthest wrong, and what changed when the reader stopped rounding. **Scope:** `examples/crystallography/6_oracle/proof_positive_control.py`, catalog CRY-6-001, run 2026-09-09

Note, 26 September. `proof_positive_control.py` imports `measure.shift_agreement`, `representation.exact` and `representation.structure.crystal` from `src/engine/python/`, which commit 74601c1 deleted on 26 September; none is in a file at orior 1948ae1. Every number below is from the run of 9 September, before the deletion. **Later, 26 September.** Commit d09b489 (orior PR 12) put back the 126 files 74601c1 deleted, each byte-identical to its state before the deletion, and biohub pins orior at d09b489. Every file named here that 74601c1 deleted is at the pin again; nothing here was rerun against them.

3.1 WHAT WAS SWEPT

Sixty four mineral names were put to the Crystallography Open Database by text search. Every entry returned with right angles was fetched, tiled, and asked for its period along each of its three axes.

names searched	64
minerals that returned a usable entry	55
structures read	151
axes measured	453

One structure contributes up to three axes, and every axis is an independent question carrying its own published answer. A wrong result cannot hide in a small denominator, which a sweep buys and a sample does not. The first version of this control took three minerals and returned three exact answers, and a coincidence still lives at three.

The same corpus was read twice, and the difference between the two readings is the subject of this chapter. The first reading put every site on a grid of 0.25 angstroms before the detector saw it. The second carried each site as an exact integer and rounded nothing.

3.2 THE RESULT

axes equal to the published edge	453 of 453, 100.0 percent
error	zero, on every axis

tolerance applied	none
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Equal as integers, with no tolerance. Every coordinate is carried at 10^{-1024} angstroms, a period is read off the whole difference set of the axis, and the comparison against the published edge is an equality between two integers. There is no tolerance to set and no error to average, because there is no error.

The failure stated in advance was any axis whose recovered period was not equal to its published edge. A near miss would have counted as a miss, since nothing in this path rounds. No axis missed.

3.3 THE READING THIS REPLACED, AND WHAT IT WAS MEASURING

The same 453 axes, read on the grid:

axes inside one voxel of 0.25 angstroms	453 of 453, 100.0 percent
axes equal to the published edge	0 of 453
mean absolute error	0.0124 angstroms
median absolute error	0.0108 angstroms
90th percentile	0.0243 angstroms
99th percentile	0.0404 angstroms
worst single axis	0.0554 angstroms

Read alone, that is a good result. Read beside the exact one, it says something else: the grid reading never once landed on the published edge, and every figure in the table is the grid.

None of that error was in the deposit and none of it was in the detector. A cell edge is published to five decimal places and the detector compares integers. The 0.0124 was introduced by this work, in the reader, before the measurement began, and it was then reported as the accuracy of the instrument.

3.4 THE BIAS, AND ITS CAUSE IS NOW SETTLED

The grid reading carried a bias:

mean signed error	-0.0067 angstroms
axes read long	120
axes read short	327

The detector read short about three times as often as it read long, and the mean sat below zero by roughly half the median absolute error. A symmetric error would have put those counts near even.

An earlier version of this chapter named the voxel floor as the candidate explanation, predicted a bias of about half a voxel spread over the tiling, and said plainly that it had not been tested by re-running with a different rounding. The bias was a measured fact with no mechanism attached.

It has now been tested, by removing the rounding. The exact reading of the same 453 axes has a signed error of zero, 0 axes long and 0 axes short. The bias was the voxel floor in its entirety. Nothing of it survives when sites are not assigned to cells.

That is the strongest form the test could take. Changing the rounding rule would have moved the bias and left the question of how much was rule and how much was something else. Removing the quantization removes the bias completely, which settles it.

3.5 THE FOUR WORST AXES, AND WHERE THEY ARE NOW

mineral	entry	axis	published	on the grid	grid error	exact
corundum	1544906	c	2.8446	2.9000	+0.0554	2.8446
herzenbergite	9008295	a	4.1480	4.1000	−0.0480	4.1480
quartz	4339381	c	15.4780	15.5250	+0.0470	15.4780
herzenbergite	9008785	b	11.1800	11.1389	−0.0411	11.1800

Herzenbergite appears twice, and it is the mineral that failed outright before the harmonic family was capped at two members. It was the hardest case in the set on the grid. It is exact on both entries now, and so is every other axis. The concept of a worst axis no longer applies to this measurement.

The two largest grid errors had opposite signs, which was worth noting when the grid was the reading: the worst case was not the bias but the resolution of the straddling ratio, and those were separate effects. Both are properties of a quantization that is no longer performed.

3.6 WHAT THIS TABLE IS NOT

It is not a measurement of any of these minerals. Every published edge in the table was measured by somebody else, to a precision this does not reach, and is the input the recovered value is graded against. The exact reading recovers the deposited number and makes no claim beyond it: agreeing with a deposit to the digit is not the same as the deposit being right, and the bracketed uncertainty a deposit carries is a quantum fixed upstream by the people who measured it.

It is also not 453 independent structures. It is 453 axes over 151 structures over 55 minerals, and several minerals contribute many entries. Treating the 453 as independent samples of anything would overstate what a sweep of 55 minerals shows.

WHERE THE PRECISION COMES FROM

Purpose: Explain how an answer landed finer than the grid it was measured on, and why that question stopped needing an answer. **Scope:** `recover_lattice_period` and `recover_exact_period` in `src/engine/python/measure/shift_agreement.py`

Note, 26 September. `src/engine/python/measure/shift_agreement.py` was deleted with `src/engine/python/` by commit 74601c1 on 26 September, and `recover_lattice_period` and `recover_exact_period` are defined in no file at orior 1948ae1; the examples that call them import the deleted tree. This chapter describes them as they were before the deletion. **Later, 26 September.** Commit d09b489 (orior PR 12) put back the 126 files 74601c1 deleted, each byte-identical to its state before the deletion, and biohub pins orior at d09b489. Every file named here that 74601c1 deleted is at the pin again; nothing here was rerun against them.

4.1 THE OBJECTION

The grid was 0.25 angstroms. Errors below that were reported. A reader should stop there and ask how a measurement resolves anything finer than the spacing it samples at, because most of the time the answer is that it does not and the extra digits are decoration.

That objection was correct to raise and it had an answer. The answer is kept below, because the mechanism is real and applies to any measurement taken on a grid. What follows it is the better resolution, which is that the grid was never necessary.

4.2 QUANTIZATION MOVES THE PERIOD, IT DOES NOT DESTROY IT

A cell edge is almost never a whole number of voxels. On a 4.148 angstrom axis at 0.25 the period is 16.59 voxels. Successive tiles therefore do not land on the same lag: they land alternately on 16 and on 17, and the share of the agreement sitting at the upper lag carries the fraction between them.

So the quantity is not lost at quantization, it is moved into the ratio between two adjacent lags. The recovered edge is the lag plus that fraction, times the voxel. Nothing is interpolated between samples that were never taken. Both lags were measured, and their relative weight is a measured quantity too.

This is why tiling mattered beyond having more than one period. The fraction is only readable because many tiles are present and their offsets accumulate. One period gives one lag and no ratio.

Measured, that recovery worked. Three cells at 10.1000, 8.6633 and 14.0574 angstroms came back at 10.1018, 8.6545 and 14.0568, and across the whole sweep 453 axes landed inside a voxel at a median absolute error of 0.0108.

4.3 THE BETTER ANSWER, WHICH IS THAT THE QUESTION WAS OURS

Everything above answers a question this work created. The voxel came from the reader in this tree and not from the deposit: a CIF publishes a cell edge to five decimal places, and the grid was imposed before the detector ever saw it.

Carry the sites as exact integers instead and the objection dissolves. It is not answered. There is no sample spacing to beat, no straddling lags, no fraction to resolve, and no error distribution to report. The same 453 axes come back equal to their published edges.

The lesson generalizes past crystals. The chapter keeps both halves for that reason. An instrument that answers finer than its own resolution deserves the scrutiny this chapter opens with. Sometimes the mechanism is sound, as it was here. It is still worth asking first whether the resolution was anybody's to impose.

4.4 WHAT THIS DOES NOT LICENSE

It does not license a claim that a cell edge is now known to a thousand digits. The exact reading recovers the number the deposit carries and stops there. A deposited edge written as 4.76050(5) has an uncertainty of five in its last place, fixed upstream by the people who measured the crystal, and dropping that bracket is all this work does with it.

What the exactness buys is that no second uncertainty is added underneath the first. The instrument contributes nothing to the error. Whatever error remains belongs to the deposit and can be argued about on its own terms.

WHERE IT STOPS

Purpose: State the boundary of the claim. Nothing here is quoted past what it supports. **Scope:** the whole chapter set.

5.1 RIGHT ANGLES ONLY

Every entry read has a cell with right angles. Oblique cells are skipped, not measured and reported as failures. A shift along a grid index is a shift along a crystal axis only when the axes are orthogonal, and on an oblique cell the grid would have to be sheared first.

That is a real restriction on coverage. It restricts which crystals were read and claims nothing about which crystals could be. Whether the same detector reads a monoclinic or triclinic cell after a shear is untested here, and the honest answer is that nobody has tried it.

5.2 TWO ATOMS ARE TWO ATOMS

This section used to record a limit that no longer exists. At 0.25 angstroms two atoms could land on the same voxel and the later one overwrote the earlier, with nothing recovering the lost site. It was harmless for the lattice period, since a period is a property of the repeating arrangement and not of any particular site, but it meant the grid was not the structure.

Exact coordinates collide only where a deposit puts two sites at the same place. Nothing is lost that the deposit did not already carry. The limit was the reader's and it is gone with the reader that had it.

5.3 EXACT DOES NOT MEAN TRUE

The exact reading recovers the number a deposit carries and stops there. A cell edge written as 4.76050(5) has an uncertainty of five in its last place, fixed by the people who measured the crystal, and dropping that bracket is all this work does with it. Agreeing with a deposit to the digit is agreement with a deposit.

What the exactness rules out is a second uncertainty underneath the first. The instrument contributes nothing to the error. Whatever error remains is the deposit's and can be argued about on its own terms.

5.4 THE DETECTOR IS NOT THE MEASURE

This work carries two instruments and reporting one as the other has caused most of the confusion it has had to correct. The crystal result belongs to the shift agreement detector, the same one that returns an image width from a byte sequence and a key length from a Vigenère cipher. It is the instrument that has external ground truth.

The null permutation identity, which carries most of the findings elsewhere in this work, has no external positive control and is not what read these cells. A crystal row does not transfer to it. Anyone quoting the crystal agreement as evidence that the departure measure works has moved a result between two instruments that share nothing but a repository.

5.5 WHAT A CELL EDGE IS WORTH HERE

Nothing to crystallography. The cells were published to more decimal places than this reaches, by refinement procedures built for the purpose, and a cell recovered from a deposited cell is a round trip and not a new measurement.

What it is worth is the direction of the evidence. An instrument that finds a period nobody told it about, on hundreds of axes whose answers were written down in advance by other people, has been shown to find structure that is there. Every control before it could only show the instrument does not invent structure that is not. Those are different claims and only one of them can be earned with a memoryless control.

5.6 THE OPEN QUESTION THIS LEAVES

The detector reads a periodicity. A crystal is the case where periodicity is exact and known. It is the easiest possible instance and it says nothing about the hard one: how far the period has to degrade before the detector stops finding it. Disorder, thermal displacement, a superlattice, an incommensurate modulation and a quasicrystal are all cases where the answer is interesting and none of them was measured.

That is the experiment worth running next, and it is worth running against published structures where the degree of disorder is itself published, which keeps the answer somebody else's number.

BIBLIOGRAPHY

Every measurement in this work is built on somebody's published result, and the first purpose of this chapter is that those people are credited by name. A surname in a footnote is not a citation. The second purpose is that the credit is checkable: each source is registered in `orior_citations`, where the copy sits itself, and this bibliography is generated from that registry and pinned to one commit of it. A reference therefore resolves not to a title but to exact bytes.

Credit here does not depend on agreement. Where this work reproduces a published result it says so, and where it fails to reproduce one it says that too, under the same names, because a refutation rests on the original as much as a confirmation does.

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[1] Crystallography Open Database. <https://www.crystallography.net/cod/>.